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Thermal characterizations of the paraffin wax/low density polyethylene blends as a solid fuel

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ABSTRACT

Thermal characterizations of a novel solid fuel for hybrid rocket application, based on the paraffin wax blends with low density polyethylene (LDPE) concentration of 5% (SF-5) and 10% (SF-10) were conducted. Both the increased regression rate in comparison with the polymeric fuel, and the improved combustion efficiency in comparison with the pure paraffin fuel reveal that the blend fuels achieve higher combustion performance. The morphology of the shape stabilized paraffin wax/LDPE blends was characterized by the scanning electron microscopy (SEM). Although the SEM observation indicated the blends have uniform mixtures, they showed two degradation steps confirming the immiscibility of components in the crystalline phase from thermogravimetric analysis (TGA). The differential scanning calorimeter (DSC) results showed that the melting temperature of LDPE in the blends decreased with an increase of paraffin wax content. The decreasing total specific melting enthalpy of blended fuels with decreasing paraffin wax content is in fairly good agreement with the additive rule. In thermomechanical analysis (TMA), the linear coefficient of thermal expansion (LCTE) seems to decrease with an increase of LDPE loading, however, the loaded LDPE do merely affect the LCTE in case of the blends with low LDPE concentration. It was found that a blend of low concentration of LDPE with a relatively high concentration of paraffin wax can lead to a potential novel fuel for rocket application, a contrary case with respect to the field of phase change materials (PCM) where a blend of high concentration of LDPE is usually used with low concentration of paraffin wax.

1. Introduction

Paraffin waxes, usually known as fuels for candle, are mixtures of saturated hydrocarbon composed of numerous alkanes. Because of their high latent heat, low vapor pressure in the melt, chemically inert and low acquisition cost (without mentioning all their positive characteristics), paraffin wax has been one of the best candidates for the phase change materials (PCM) for thermal storage applications and electronic cooling devices [1–4]. In the other hand, hydrocarbon polymers releasing a lot of thermal energy are widely used as fuels in various industrial areas [5–7]. In particular, polymers such as hydroxyl-terminated polybutadiene (HTPB), polyethylene (PE), polymethylmethacrylate (PMMA) and polypropylene (PP) have shown potential possibilities as solid fuels for hybrid rocket motor (HRM) [8–10] in aerospace industry. However, these typical polymers have relatively slower burning speed with respect to the composite fuels which are widely used in solid rocket system provoking low thrust performance. Typically, this slow burning speed is associated to the intrinsic combustion mechanism of the hybrid rocket. The burning speed known as regression rate is defined as the rate the fuel surface recedes and it is a key parameter for the hybrid rocket motor design. Therefore, a lot of research efforts [10–13] have been conducted to enhance the fuel regression rate.

Karabeyoglu et al. [14] have suggested that paraffin wax can be a promising solid fuel for overcoming the low regression rate of HRM fuel because its regression rate has been found to be 3–4 times higher than that of conventional polymeric fuels. Fast regression rate of these paraffin-based fuels results from the important phenomenon of two-phase mass transfer by the gas and liquid, which is different from gas phase mass transfer of conventional polymeric fuels.

Fig. 1 is an illustration of two-phase mass transfer mechanism by the paraffin wax from the fuel surface. First, the heated paraffin wax fuel forming a melted liquid layer over the solid fuel surface generates unstable wavelets in which some portions are vaporized to the gas phase. Secondly, the substantial liquid fuel droplets which are generated at the tips of the wavelets are entrained into

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Nomenclature

A_t Nozzle throat area (m^2)
c^*_{exp} Experimental characteristic velocity (m/s)
c^*_{theo} Theoretical characteristic velocity (m/s)
h Melt layer thickness (m)
$\dot{m}_{ent}$ Entrained fuel mass flux from fuel surface (kg/m^2 s)
$\overline{\dot{m}}_o$ Time averaged oxidizer mass flow rate (kg/s)
P_c Chamber pressure (Pa)
P_d Dynamic pressure (Pa)
t_b Burning time (s)
α Dynamic pressure exponent
β Thickness exponent
γ Viscosity exponent
η_{c*} Efficiency of characteristic velocity
Δm_f Mass difference between initial and final fuel grain (kg)
μ_l Liquid layer viscosity (Pa s)
π Surface tension exponent
σ Surface tension (N/m)

the gas stream where this entrained mass of the liquid droplets is known to be a dominant mechanism leading to the increase of the regression rate.

However, manufacturing of large fuel grain using paraffin-based fuel is a difficult task. Kilic et al. [15] have reported that poor mechanical characteristics and low melting temperature of the pure paraffin fuel can lead to a serious deformation under storage, handling or operating conditions. In their study of large fuel grain for space shuttle booster, fuel storage life time was limited due to the slump deformation for temperature range above 45 °C. In addition, combustion efficiency of the paraffin fuel is lower than the usual polymeric fuel because fuel droplets generated from the melted liquid layer are not completely burned during the passage of the fuel grain port and through the exhaust nozzle [16,17]. In many reports and papers [12,18–21], use of heterogeneous materials are presented where energetic metal particles and carbon black powders were added to pure paraffin wax to improve the mechanical strength, combustion efficiency and fuel density. Nevertheless, adding such additives to paraffin wax has often been a problem since it occasionally leads to combustion instability [21–23], chamber pressure sensitivity and non-uniform concentration of added material during mixing and casting process.

In this respect, this paper suggests an effective solid fuel by blending pure paraffin wax and polyethylene with different objective of PCM application. The motivation of this research is due to the following reasons. Firstly, paraffin-LDPE blended fuels may have many potential advantages compared to pure paraffin wax which includes additives. The fact that both paraffin wax and LDPE can be considered as a series of homologous materials, the proposed blended fuel may behave as a uniform material if mixed and prepared properly. Secondly, paraffin-LDPE blended fuels can improve the mechanical strength and the combustion efficiency of pure paraffin fuel since the physical properties of blended fuel are better than the paraffin itself [24].

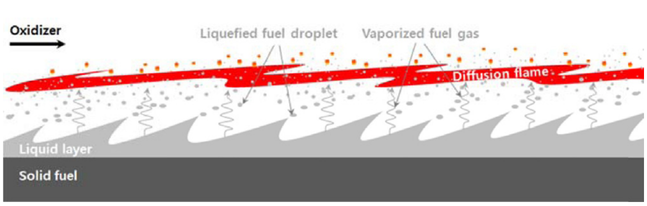

Fig. 1. Combustion mechanism of paraffin-based liquefying solid fuel.

Many researchers have already been focused on the paraffin wax/LDPE blend solution which is one of the best candidates as a phase change materials (PCM) for thermal storage applications [1–4]. Usually, the LDPE containing range in the blends for PCM application [1] is over 40%.

In our previous study [24] on the mechanical strength and combustion performance of blends, an entirely different range from that of blends for PCM has been tried. The paraffin wax blends containing less than 10% of LDPE have been found to be very effective for hybrid rocket fuel application with sufficient burn rate. Consequently, understanding the thermal characterizations of blends containing less than 10% of LDPE would be necessary to investigate.

In this paper, to evaluate good potentiality required for hybrid rocket application, paraffin wax blends with low density polyethylene were prepared and tested. The propulsion performances based on the solid fuel regression rate and combustion efficiency were compared by burning tests using lab-scale hybrid combustor. In order to examine the uniformity of the new blends fuel, the structure of blends was investigated using scanning electron microscopy (SEM). Thermal properties, such as crystallization temperature, melting temperature and specific melting enthalpy obtained from the differential scanning calorimeter (DSC) were discussed to describe the behavior of liquid layer. During the thermogravimetric analysis (TGA), decomposition temperature was obtained to determine the onsets of drastic thermal history while the thermal stability is analyzed to highlight the degradation behavior. Also, in order to consider a possible thermal deformation, thermal expansion behavior is examined by the thermomechanical analysis (TMA). In the present paper, thermal properties of paraffin wax/LDPE blends with low LDPE content are investigated and discussed especially for the standpoint of hybrid rocket fuel application.

2. Paraffin wax/LDPE blends as hybrid rocket fuel

2.1. Effective LDPE concentration

The conventional HRMs storing the solid fuel and liquid oxidizer separately gain their thermal energy from the turbulent diffusion combustion mechanism occurring in the boundary layer over the solid fuel surface. Usually the heat transfer from the flame zone to the fuel surface is transferred by convection and radiation, and leads to vaporization of solid fuel. At this point, slow regression rate of HRM fuel is due to the decreased heat transfer by the blocking effect which is triggered by the radial blowing of decomposed fuel from the solid fuel surface to the flame zone [25].

As mentioned before, Karabeyoglu et al. [14] showed that the paraffin wax can be an effective solid fuel to innovatively enhance the low regression rate of HRM fuel. This high regression rate of paraffin wax fuel is achieved by not only gasified mass transfer as in conventional polymeric fuels, but also additional mass transfer by shear driven liquid droplets.

In order to model and scale the entrainment rate of liquid droplets, Karabeyoglu et al. [14] suggested an empirical formula (Eq. (1)) which includes the gas dynamic pressure in the combustion chamber, the melt layer thickness, the surface tension and the viscosity of the liquefied paraffin at the fuel surface.

$$\dot{m}_{ent} \propto \frac{{P_d}^{\alpha} h^{\beta}}{{\mu_l}^{\gamma} \sigma^{\pi}} \tag{1}$$

Eq. (1) shows that the entrained droplets mass transfer mechanism from the fuel surface depends not only on the operational conditions above the melted fuel surface but also on the material properties beneath the melted surface, indicating the ways how to increase the regression rate.

Eq. (1) shows that a paraffin wax based blends or mixtures with different values of surface tension and viscosity of the melt layer can play a role for controlling the regression rate and the mechanical strength of the fuel. For example, the augmented surface tension and viscosity by adding LDPE will decrease the entrained droplets that are usually not completely burned during the passage of the fuel grain. Simultaneously, introduction of LDPE into the paraffin wax matrix with short chains will lead to the reinforcement of the mechanical strength.

In this respect, we have found in our previous study [24] that paraffin wax based blends containing less than 10% of LDPE were very effective for enhancing the combustion efficiency and mechanical strength with sufficient burn rate.

For more description, some of our previous test results [24] are shown in Fig. 2. Fig. 2 shows the comparisons of test results of mechanical strength of the paraffin wax with the two blends, each containing 5 wt% of LDPE and 10 wt% of LDPE.

One can see that the mechanical strengths can be improved as the LDPE concentration is increased in the paraffin wax and that a small LDPE contents (5 wt%, 10 wt%) in the blend is enough to increase the tensile and compression strength. The results reveal that the tensile and compressive strength of SF-10 having LDPE portion of 10 wt% show a strength increment of 42.4% and 42.2%, respectively, compared to those of pure paraffin. This result clearly demonstrates that mechanical strength of blends is sufficient to prevent the structural deformation by adding a small amount of LDPE compound to the pure paraffin.

For the combustion performance, more extended burning tests on the regression rate and combustion efficiency are conducted in this work. These comparison results will be presented in Section 4.

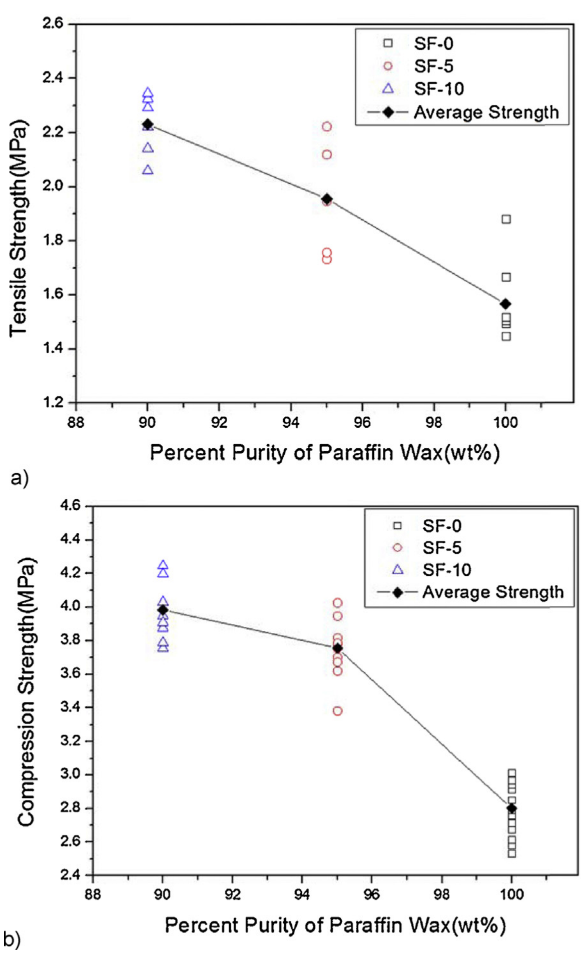

Fig. 2. Mechanical strength of paraffin-based fuels [24]. (a) Tensile strength of paraffin based fuels w.r.t. pure paraffin percentage. (b) Compression strength of paraffin based fuels w.r.t. pure paraffin percentage.

3. Experimental

3.1. Materials

In this investigation, commercially available products were used for the paraffin wax and LDPE. A fully refined soft paraffin wax of carbon number 28 supplied by Nippon Seiro Co., Ltd., was used with melting point of 61 °C, molecular weight of 394 g/mol and density of 0.916 g/cm^3 at 25 °C. The LDPE supplied as pellets by Hanwha Chemical Co., Ltd., was used for blending material with melting point of 110 °C, molecular weight of 3.0×10^5 g/mol, density of 0.921 g/cm^3, melt flow index (MFI) of 0.3 g/10 min at 200 °C and 2.16 kg. These two products will be the base materials for the blending process.

3.2. Preparation of solid fuel samples

Generally, samples of polymer blends are made by mixing components in a heated screw extruder or mechanically mixing and molding through the compression. Considering that the practical purpose of the paraffin wax/LDPE blends used in this study is for solid fuel application, blends samples were prepared by melt compounding to simulate similar conditions for the actual fuel manufacturing process of HRM. The wax was first prepared in a mixing and melting chamber on the magnetic hotplate stirrer. Afterward, the LDPE pellets were added to molten wax at 150 °C. The paraffin wax/LDPE mixture was blended by magnetic bar for 10 min at a mixing rate of 30 rpm. The molten blends were then poured into a mold and cooled at room temperature for over 4 h. Sample blends used in this study are listed in Table 1.

3.3. Methods used for blends characterization

The burning tests using lab-scale hybrid motor were conducted for SF-0, SF-5 and SF-10 solid fuels to highlight the combustion performance of the proposed blends fuels. The used experimental setup is mainly composed of oxidizer feed system, ignition system, data acquisition system and hybrid rocket motor using cylindrical grain with a single fuel port. Detail experimental setup and test procedure were already described in our previous paper [24]. The specifications of fuel grain configuration and test conditions for this study are shown in Table 2.

SEM observations were performed for SF-0, SF-5, SF-10 and SF-50 blends to check the morphological traits when the LDPE concentrations are different. The samples were cracked at room temperature and the fracture surfaces have been observed. SEM analyses were carried out using a Hitachi S-4700 microscope where the acceleration voltage was set to 5.0 keV. The samples'

Table 1
Identification and compositions of the solid fuels.

Solid fuel	Compositions
SF-0	Paraffin wax 100%
SF-5	Paraffin wax 95% + LDPE 5%
SF-10	Paraffin wax 90% + LDPE 10%
SF-100	LDPE 100%

Table 2
Specifications of burning test conditions for the cylindrical hybrid motor.

Solid fuel	SF-0	SF-5	SF-10
Averaged oxidizer mass flux($kg/m^2 \cdot s$)	52–152	69–286	82–289
Oxidizer	Gaseous oxygen		
Burning time (s)	5		
Fuel outer diameter (mm)	70		
Fuel port length (mm)	100		
Fuel port diameter (mm)	10		
Fuel length to diameter ratio	10		

fracture surfaces were coated with platinum and rhodium by an electro-deposition method to impart electrical conduction before analyzing the SEM images.

DSC measurements were carried out in a PerkinElmer Pyris 1 DSC thermal analyzer. Samples were heated from 20 to 200 °C at a heating rate of 10 °C/min under nitrogen atmosphere and then cooled at a cooling rate of 10 °C/min. Thermal properties, such as melting temperature and heat flow were determined from the second heating results where the mass of each sample was in the range 2–4 mg.

TGA was performed by a TA Instrument 2950 TGA from 25 to 800 °C at a heating rate of 10 °C/min in a flowing nitrogen. The mass of each sample was about 10–15 mg in TGA. For each of these TGA measurements, DSC analyses were repeated three times consecutively and the results were reproducible within ±5%.

TMA was also conducted using a Shimadzu TA-50 thermo mechanical analyzer. The fuel samples were measured over a temperature range of −40 to 60 °C at a scanning rate of 5 °C/min under argon. A plot of displacement depending on the temperature was recorded. Then, the thermal expansion coefficient was obtained from the slope of the thermogram.

4. Result and discussion

4.1. Combustion performance comparisons between solid fuels

Fig. 3 compares measured regression rates of all test cases conducted in this study with SF-100 and HTPB data [24]. One can notice that the regression rates of all liquefying paraffin-based fuel (SF-0, SF-5, SF-10) are highly increased than those of non-liquefying polymeric fuel (SF-100, HTPB), and that the regression rate of liquefying fuels decreases as LDPE wt% increases. It is believed that the relatively low regression rate of SF-5 and SF-10 with respect to SF-0 would be due to the decrease of entrainment regression rate since the fuel mass entrained at the liquid-gas interface is usually diminished as the viscosity and surface tension increase. Thus, it can be regarded that this rheological property difference leads to the regression rate difference. Although the regression rates of each blended fuels (SF-5, SF-10) are beneath the value of pure paraffin (SF-0), they are still much higher than that of HTPB which is the most popular solid fuel used in hybrid rocket motor application.

Fig. 4 compares the combustion efficiency with respect to the total propellant mass flow rate. Combustion efficiency based on characteristic velocity was used to evaluate the performance of various fuels. Efficiency of characteristic velocity can be calculated by using the following equations:

$$c^*_{\mathrm{exp}} = \frac{A_t \int_0^{t_b} P_c \mathrm{d}t}{\int_0^{t_b} \dot{m}_o \mathrm{d}t + \Delta m_f} \tag{2}$$

$$\eta_{c^*} = \frac{c^*_{\mathrm{exp}}}{c^*_{\mathrm{theo}}} \tag{3}$$

where the theoretical characteristic velocity was calculated by CEA [26] code. As can be observed, a 20% augmentation of the combustion efficiency is achieved with the blended fuels (filled circle and triangle) containing less than 10% of LDPE with respect to the pure paraffin wax fuel (filled square). It has been concluded that, even with a small wt%, the LDPE can be an effective mixing ingredient which can improve the combustion efficiency by regulating the LDPE concentration in the paraffin wax fuel. Although the combustion efficiency of SF-100 polymeric fuel shows the highest value, it is hard to be an effective fuel for hybrid application. This is because its low regression rate basically produces low thrust level.

4.2. Morphology comparison of solid fuels

The SEM images of pure paraffin and paraffin wax/LDPE blends at different compositions are shown in Fig. 5. These images clearly depict the differences in the morphology of paraffin-based blends with respect to the LDPE mixture ratio. For example, Fig. 5d representing the blend containing 50% LDPE shows that its morphology is distinct with respect to three other cases which have low LDPE fraction. This image demonstrates large LDPE aggregates (arrow A in Fig. 5d) and phase separation between LDPE and paraffin wax. It is, therefore, predictable that interfacial adhesion between paraffin wax and LDPE would be weak. Krupa

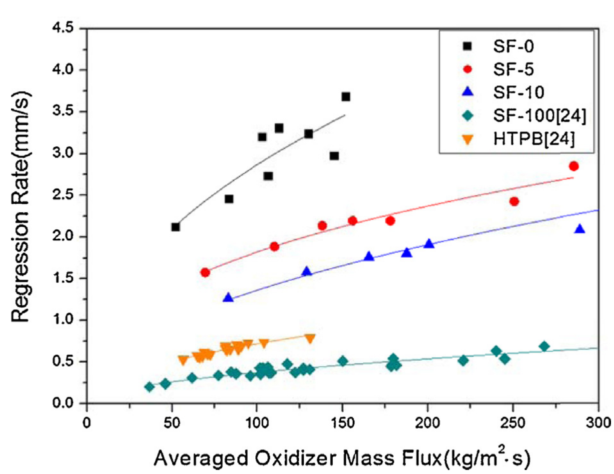

Fig. 3. Regression rates for paraffin wax/LDPE blends (SF-5, SF-10) and pure components (SF-0, SF-100, HTPB).

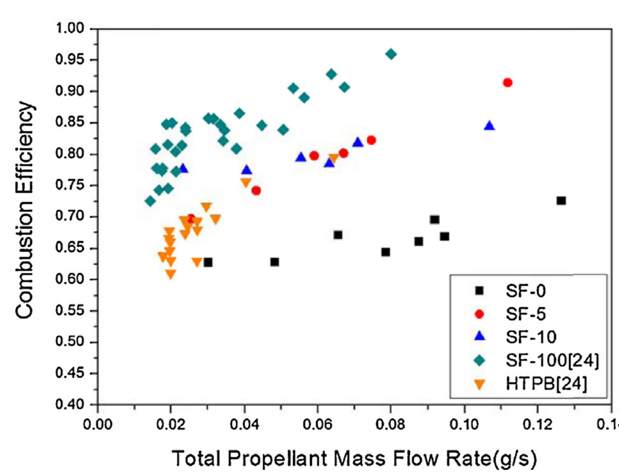

Fig. 4. Combustion efficiency for paraffin wax/LDPE blends (SF-5, SF-10) and pure components (SF-0, SF-100, HTPB).

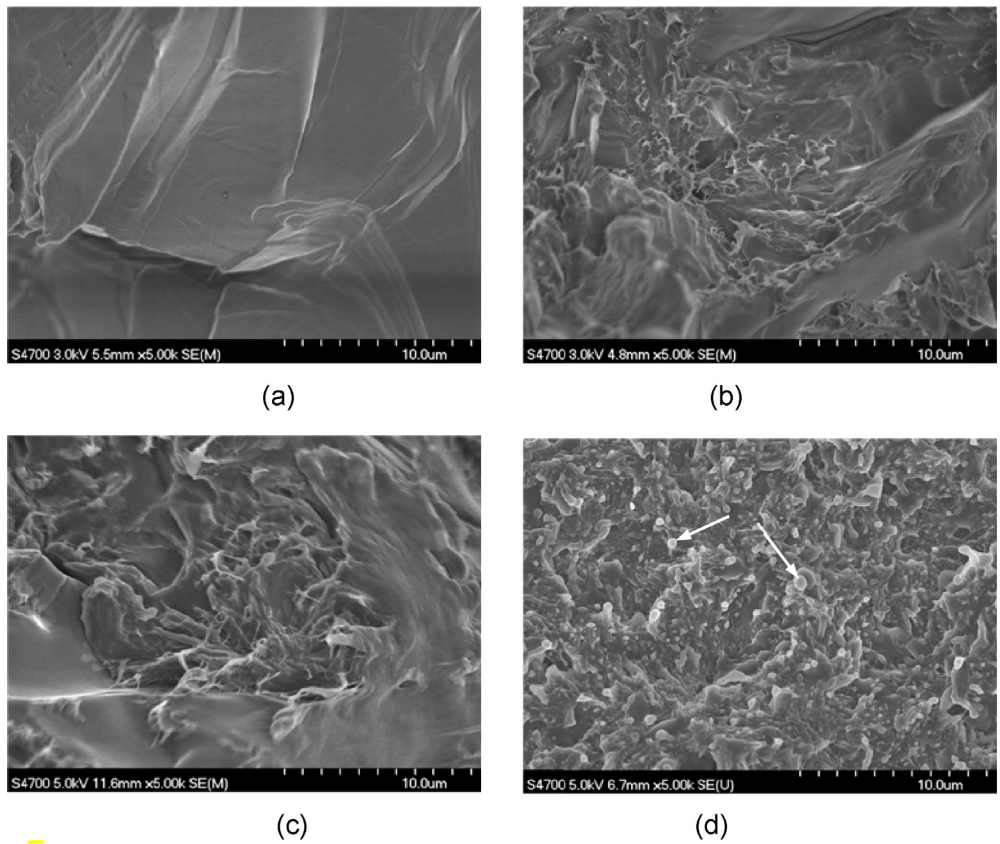

Fig. 5. SEM images of (a) 100/0 w/w paraffin wax/LDPE (SF-0), (b) 95/5 w/w paraffin wax/LDPE (SF-5), (c) 90/10 w/w paraffin wax/LDPE (SF-10) and (d) 50/50 w/w paraffin wax/LDPE (SF-50).

et al. [1] explained that this behavior is due to the difference of molecular weight and of structure between LDPE and paraffin wax in their recent publication. Particularly, they found that the soft paraffin wax having a low molecular weight and low viscosity can be easily separated from the blends of paraffin wax/LDPE of 50% each. On the other hand, for Fig. 5b and c whose LDPE ratio is low, the LDPE is well dispersed showing a fairly homogeneous surface without separation within the paraffin wax compound. This morphology implies that a macroscopic homogeneity of the blends may be achieved below a certain percentage of LDPE content in the blend mixture. From these observations, the SF-5 and SF-10 blends can be regarded as a uniform-like mixture at least at the macroscopic level. If this is the case, the liquid droplets produced from the SF-5 or SF-10 blends during the burning time can be assumed as a single ingredient where combustion analysis and modeling would be easier and simpler. The following DSC measurement can clarify the nature of the homogeneity level of the wax/LDPE blends.

4.3. Differential scanning calorimetry

Fig. 6 shows the DSC heating curves of pure materials (SF-0, SF-100) and of blends (SF-5, SF-10) used in this study. The DSC curve of pure paraffin wax (solid line) shows clearly two endothermic peaks at 45.3 °C and 61.2 °C. The first peak relates to the solid–solid transition which is the transition of one crystalline phase to another. The second peak is associated with the melting of the crystallites. These two distinct peaks are also observed with the two blends fuels (SF-5, SF-10) at the same temperature, but to a lower level of heat flow compared to SF-0. One can see that the curves corresponding to the blends are fairly similar to each other, and they do not show any notable difference within the LDPE fraction between 5 and 10%. Furthermore, although low in magnitude, the heating curves of these two blends show one additional endothermic peak related to the melting of LDPE. The DSC results are summarized in Table 3. The melting temperature of pure LDPE was found to be 108.6 °C whereas, 94.7 and 93.0 °C were measured for the melting temperature of SF-10 and SF-5, respectively. While the melting temperatures ($T_{m,2}$) of paraffin based fuel do not show any converging behavior, the melting temperature of LDPE ($T_{m,3}$) decreases with an increase of paraffin wax content. These trends are caused by the formation of smaller

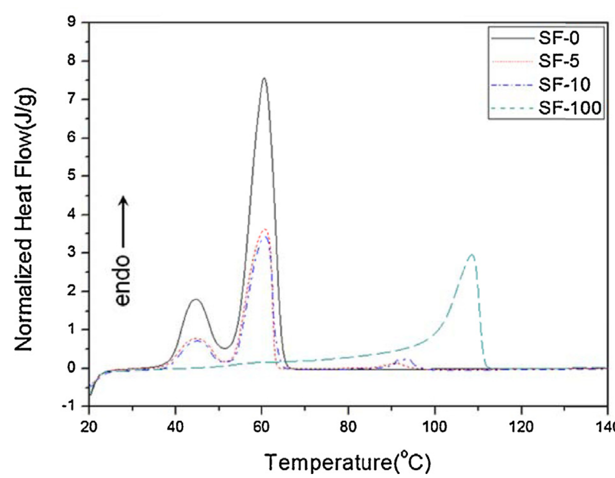

Fig. 6. DSC heating curves for paraffin wax/LDPE blends (SF-5, SF-10) and pure components (SF-0, SF-100).

Table 3
Parameters from DSC measurement of paraffin wax/LDPE blends (SF-5, SF-10) and pure components (SF-0, SF-100).

Solid fuel (wax/LDPE)	$T_{m,1}$ (°C)	$T_{m,2}$ (°C)	$T_{m,3}$ (°C)	$\Delta h_{m,2}$ (J/g)	$\Delta h_{m,3}$ (J/g)	$\Delta h_{m,tot}$ (J/g)	$\Delta h^{(4)}_{m,tot}$ [a] (J/g)	EER [b] (%)	$T_{c,1}$ (°C)	$T_{c,2}$ (°C)	$T_{c,3}$ (°C)	$\Delta h_{c,2}$ (J/g)	$\Delta h_{c,3}$ (J/g)	$\Delta h_{m,tot}$ (J/g)
SF-0 (100/0)	45.3	61.2	–	193.0	–	193.0	193.0	–	41.3	57.2	–	190.1	–	190.1
SF-5 (95/5)	44.9	60.3	93.0	179.5	4.8	184.3	189.1	2.5	43.9	59.0	80.2	178.5	5.2	183.7
SF-10 (90/10)	45.1	60.5	94.7	172.3	8.7	181.0	185.2	2.3	43.7	57.5	82.5	171.1	9.8	180.9
SF-100 (0/100)	–	–	108.6	–	114.7	114.7	114.7	–	–	–	95.1	–	110.9	110.9

T: temperature; Δh: specific enthalpy; m: melting; c: cooling; 1: solid–solid transition; 2: paraffin wax melting; 3: LDPE melting; tot: total.
[a] $\Delta h^{(4)}_{m,tot}$ is the total specific enthalpy obtained from the Eq. (4).
[b] $\mathrm{EER} = \left(\Delta h^{(4)}_{m,tot} - \Delta h_{m,tot}\right)/\Delta h^{(4)}_{m,tot}$.

crystallites due to the miscibility of the components in the molten state.

The total specific enthalpy of melting was evaluated by the use of a linear baseline over a wide thermal range at 20–140 °C for all fuels. The total melting enthalpy $\left(\Delta h^{(4)}_{m,tot}\right)$ increases with an increase in paraffin wax content, which is the result of the higher crystallinity of the paraffin wax. These experimentally measured melting enthalpies were compared with the calculated melting enthalpy according to the additive rule given by Eq. (4):

$$\Delta h^{(4)}_{m,tot} = w_{paraffin}\Delta h_{m,paraffin} + w_{LDPE}\Delta h_{m,LDPE} \quad (4)$$

where $w_{paraffin}$ and $\Delta h_{m,paraffin}$ are the weight fraction and specific enthalpy of melting for paraffin wax, and w_{LDPE} and $\Delta h_{m,LDPE}$, the corresponding values for LDPE. The total calculated melting enthalpy $\Delta h^{(4)}_{m,tot}$ can be obtained by adding melting enthalpy of each composition. As presented in Table 3, a difference between experimental and calculated values of total specific enthalpy can be seen for each blend fuel. However, these differences are within 2.5% error range. This indicates that there is no leakage of paraffin wax from the blends during the sample preparation despite having relatively high paraffin wax mixture fraction.

It can be observed that the total specific melting enthalpy of blends decreases with a decrease of paraffin wax content. This trend is corollary, and the reduced melting enthalpy of blends can act as a stimulant for the fast burning rate of solid fuel since the melting enthalpy of paraffin wax is higher than that of LDPE.

The cooling curves of blends and pure components are shown in Fig. 7. Three exothermic peaks, related to the solid–solid transition of the paraffin wax, crystallization of paraffin wax and crystallization of LDPE, were found for the blends. Two peaks for the paraffin wax and one single peak for the LDPE were observed as in the heating case. All temperature and crystallization enthalpy behaviors were found to follow similar trends to the heating process.

Although the blends morphology found from SEM analysis showed a uniform-like mixture for low LDPE content (10%), these DSC results indicate that the blends are not homogeneous material since distinct peaks exist at different temperature. It is evident that the blends cannot be viewed as a single ingredient. However, the thermodynamic parameters, such as crystallization temperatures, melting temperatures and their respective melting enthalpies could provide valuable inputs for the modeling of evaporation and combustion of paraffin based fuels.

4.4. Thermogravimetric analysis

In most researches for PCM application [1–4], TGA results have been focused on the functional ability of the newly suggested blends in terms of thermal stability. However, considering the practical application of this study, more important information from TGA can be drawn for the blend fuels associated with degradation behavior and decomposition temperature. Thus, in this sub-section, our major interest will be concentrated on the acquisition of thermal information for the combustion modeling of wax/PE blends.

TGA results based on the weight loss with respect to the temperature are shown in Fig. 8. The pure components (SF-0, SF-100) decompose completely in a degradation process of single step, whereas two blends (SF-5, SF-10) show two distinguishable steps.

The thermal stability of all paraffin based fuels (SF-0, SF-5, SF-10) shows a similar behavior up to the temperature around 150 °C, and they are thermally stable regardless of LDPE content. The measured results showed no char yield at temperatures higher

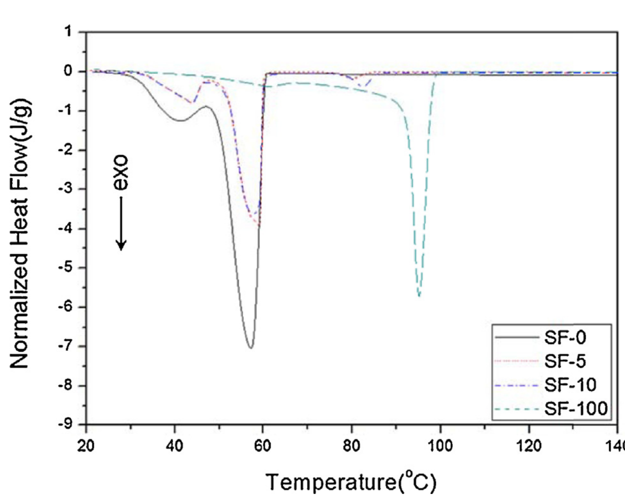

Fig. 7. DSC cooling curves for paraffin wax/LDPE blends (SF-5, SF-10) and pure components (SF-0, SF-100).

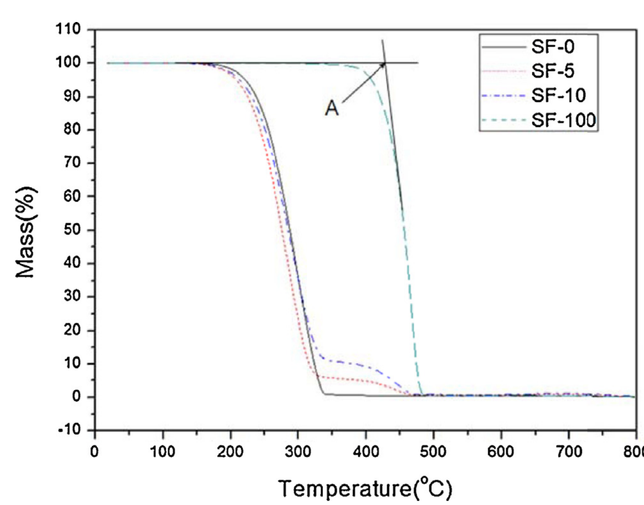

Fig. 8. TGA curves for paraffin wax/LDPE blends (SF-5, SF-10) and pure components (SF-0, SF-100).

than 800 °C. Fig. 8 confirms that the thermal stability of all paraffin based fuels is lower than that of LDPE as a consequence of the lower thermal stability of the paraffin wax as have been confirmed in Refs. [3,4]. This will be a shortcoming if the compound is to be used as a PCM, while this behavior can act as an advantage in terms of combustion efficiency in the HRM.

In order to determine the onset temperature of decomposition, we define the decomposition temperature as the temperature at the intersection of the two tangent lines (arrow A in Fig. 8). From the TGA curve of the pure wax (SF-0), one weight loss step corresponding to the decomposition temperature of wax evaporation is observed at 242.0 °C. For the SF-5 fuel, the first weight loss at temperatures around 229.7 °C is attributed to the evaporation of the pure wax while a second weight loss at temperatures around 415.6 °C is related to the decomposition of LDPE. Similar trends are observed from the TGA curve of the SF-10 fuel where the corresponding temperature for the evaporation of pure wax and decomposition of LDPE are 238.1 °C and 411.5 °C respectively. These temperatures found by the intersection method are compared to the temperature of 10% and 20% degradation (mass loss) to see the amplitude of the discrepancy where all measured decomposition temperatures associated to each fuels are summarized in Table 4.

Experimental researches [1–4] have been usually conducted to find out the decomposition behavior of blends between paraffin wax and polymer for relatively higher polymer contents. It has been reported that the decomposition temperature decreases as the paraffin wax content increases for uncross-linked blends, and that these temperatures are determined between those of pure paraffin wax and pure polymer. This is because the paraffin wax which has lower molecular weight lets sufficient energy to escape from the matrix at the low temperature [3].

On the other hand, for our blends with a very low LDPE content, decomposition temperatures and TGA curves reside in a quite similar range of the pure wax before the second weight loss. These trends did not follow the decomposition trend observed in the literatures of the higher polymer content. This indicates that the sparse LDPE distribution in the blends gives no great influence on the escape energy of the paraffin wax.

4.5. Thermomechanical analysis

The coefficients of thermal expansion were measured to evaluate the thermomechanical stability of the paraffin wax/LDPE blends, and they were compared to that of the pure paraffin wax. Fig. 9 shows the raw data of the thermal strain measured by the TMA with respect to the temperature for different LDPE content. From this result, linear coefficient of thermal expansion (LCTE) defined as the ratio of thermal strain versus temperature can be determined in a test range of −30 to 30 °C. The LCTEs obtained for the paraffin based fuels are presented in Table 5. These results indicating a trend which decreases the LCTE with increasing LDPE is, however, ambiguous. This is because the quantitative difference in LCTE values between blends and pure wax is very small. It seems

Table 4
The parameters obtained from TGA measurement of paraffin wax/LDPE blends (SF-5, SF-10) and pure components (SF-0, SF-100).

Solid fuel	$T_{d,1}$[a] (°C)	$T_{d,2}$[a] (°C)	$T_{d,10\%}$[b] (°C)	$T_{d,20\%}$[b] (°C)
SF-0	242.0	–	237.6	255.8
SF-5	229.7	415.6	224.9	243.5
SF-10	238.1	411.5	231.1	250.7
SF-100	–	429.6	420.3	434.8

[a] $T_{d,1}$ and $T_{d,2}$ are the decomposition temperature attributed to the evaporation of the paraffin wax and LDPE based on the intersection of the two tangent lines.
[b] $T_{d,10\%}$ and $T_{d,20\%}$ are the temperature of 10% and 20% degradation of the solid fuels.

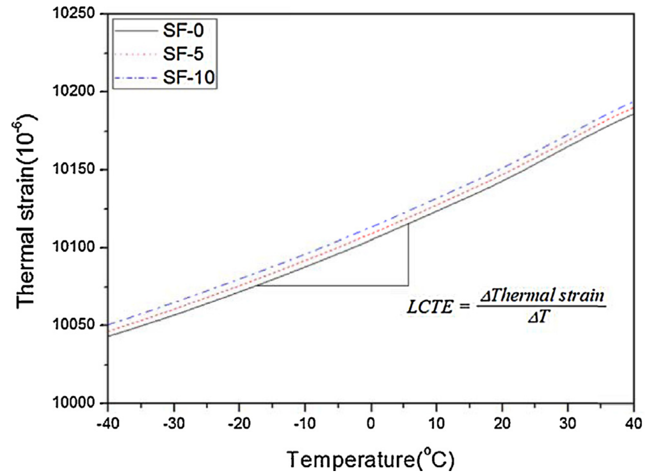

Fig. 9. Thermal strain responses of paraffin based fuel (SF-0, SF-5, SF-10).

Table 5
Values of the linear coefficient of thermal expansion of the paraffin based fuels.

Solid fuel	LCTE (10^{-6}/(°C))
SF-0	180.6
SF-5	180.5
SF-10	179.8

that the loaded LDPE do merely affect the LCTE for blends with a low LDPE fraction (<10%). Nevertheless, if we accept this trend as a meaningful observation, adding LDPE can play a positive role for the low sensitivity to the thermal deformation of fuel during handling, storage and manufacturing process of the blends fuel.

5. Conclusions

Combustion performance, thermal properties and morphology of a novel hybrid rocket fuel based on the paraffin wax blend with a low LDPE content were investigated in this paper. The burning tests using lab-scale hybrid motor were conducted for SF-0, SF-5 and SF-10 solid fuels to measure the regression rate and combustion efficiency. Thermal properties, such as crystallization temperature, melting temperature and specific melting enthalpy are obtained from the DSC results while decomposition temperature and thermal stability were analyzed by TGA to highlight the degradation behavior of the newly blended material. Thermal expansion behaviors were also examined by TMA.

The regression rates of all blended fuels are 3–4 fold higher than those of LDPE and these rates increased as LDPE wt% decreased. Although the regression rates of blended fuels are beneath the value of pure paraffin (SF-0), they are still much higher than that of HTPB. In addition, a 20% augmentation of the combustion efficiency is achieved with the blended fuels with respect to the pure paraffin wax fuel. It has been concluded that the LDPE can be an effective mixing ingredient which can improve the combustion efficiency. SEM results of SF-5 and SF-10 blends reveal that a macroscopic homogeneity of the blends may be achieved below a certain percentage of LDPE concentration, and that uniform-like mixture at the macroscopic level can be made when the LDPE content is below 10%.

From TGA analyses, the thermal stability of all paraffin based fuels (SF-0, SF-5, SF-10) shows a similar behavior up to the temperature around 150 °C, and they are thermally stable regardless of LDPE content. While the pure components (SF-0, SF-100) decompose completely in a degradation process of single step, the blends (SF-5, SF-10) show two distinguishable steps confirming the immiscibility of these components in the crystalline phase. However, DSC results confirm that the melting temperature of LDPE decreases with an increase of paraffin wax content. The total specific enthalpy of melting in the blends decreases with a decrease of paraffin wax content. These reduced melting enthalpies of blends can act as a stimulant for the fast burning rate of solid fuel and can promote the combustion efficiency.

The TMA analysis of the blends shows that the linear coefficient of thermal expansion decreases very slowly with increasing LDPE concentration. Additional tests with higher LDPE content would be interesting to see the LCTE variation with respect to LDPE concentration. Nevertheless, if we admit this LCTE decrease, adding LDPE can play a positive role for the low sensitivity to the thermal deformation of fuel during handling, storage and manufacturing process of the blends fuel.

Contrary to the phase change material application, a blend of low concentration of LDPE with a relatively high concentration of paraffin wax can lead to a potential novel fuel for hybrid rocket application.

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